

QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT - 14

A PRELIMINARY TEST

- 1 Nature of the salt : Crystalline
- 2 Colour of the salt : Colourless
- 3 Solubility:

Water
✓

Dil. HCl

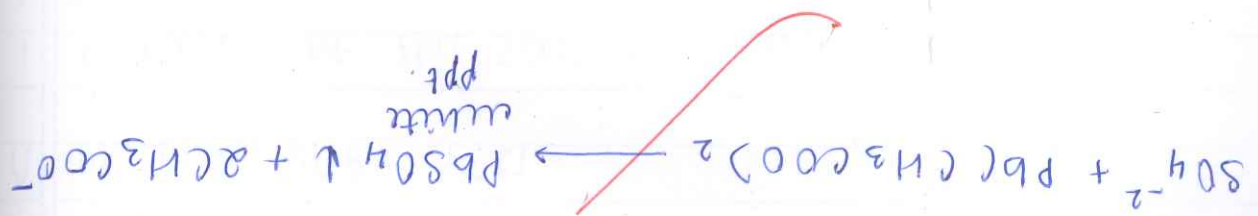
Dil. HNO₃

B Analysis of acid radicals:

Experiment	Observation	Inference
Detection of group I acid radicals.		
Salt + Dil. HCl	No observation	Group I acid radical absent

Detection of group II acid radicals:

Salt + conc. H ₂ SO ₄ in a dry test tube	No observation	
Above solution + copper turnings heated strongly.	No observation	Group II acid radical absent.



Detection of group III acid radicals (SO_4^{2-}):

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|---|--|--|
| 1. Clear solution of the salt in dil. HCl + Barium chloride solution | A white precipitate is formed. It is insoluble in excess of dil. HCl | III group acid radical is present. Sulphate is present and confirmed |
| 2. Lead acetate test: Clear solution of the salt in water + Acetic acid + Lead acetate solution | A white precipitate soluble in ammonium acetate solution | Sulphate is present and confirmed |

C Analysis of Base radicals (cations):

Test for ammonium (NH_4^+) radical (zero group analysis)

A pinch of the salt is heated with 5 drops of sodium hydroxide	No observation	NH_4^+ absent.
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Detection of group III acid radicals (SO_4^{2-}):

1. Clear solution of the salt in dil. HCl + Barium chloride solution. A white precipitate is formed. It is insoluble in excess of dil. HCl. III group acid radical is present. Sulphate is present and confirmed.
2. Lead acetate test: Clear solution of the salt in water + Acetic acid + Lead acetate solution. A white precipitate soluble in ammonium acetate solution. Sulphate is present and confirmed.

C Analysis of Base radicals (cations):

Test for ammonium (NH_4^+) radical (zero group analysis)

A pinch of the salt is heated with 5 drops of sodium hydroxide	No observation	NH_4^+ absent.
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Detection of group I basic radicals

Original solution + dil. HCl in a test tube.	No observation	I group basic radical absent
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Detection of group II basic radicals:

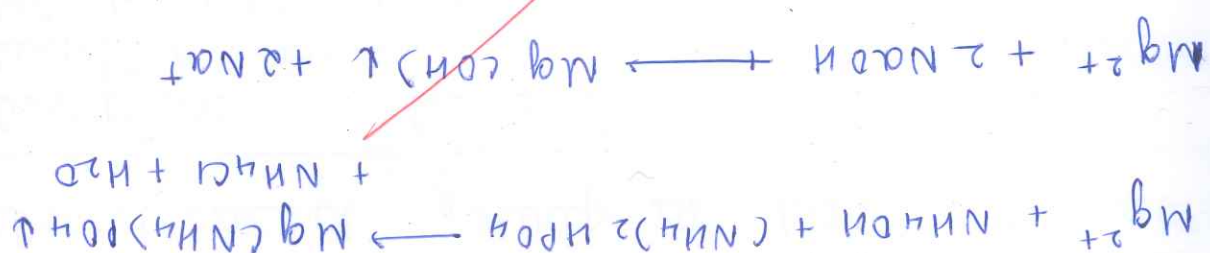
Original solution + dil. HCl + H_2S solution & added.	No observation	II group basic radical absent
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Detection of group III basic radicals:

Original solution + dil. NH_4Cl solid + NH_4OH solution in excess	No observation	III group basic radical absent.
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Detection of group IV basic radicals:

Original solution + NH_4Cl solid + NH_4OH solution in excess + H_2S	No observation	IV group basic radical absent.
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Detection of group I basic radicals:

Original solution + NH_4Cl solid + NH_4OH solution in excess + $(\text{NH}_4)_2\text{CO}_3$ solution	No observation	<u>I</u> group basic radical absent.
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Detection of group VI basic radicals :

(i) Original solution white crystalline + NH_4Cl solid + NH_4OH solution + ammonium phosphate solution.	no precipitate	Mg^{2+} is confirmed
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(ii) Original solution + NaOH solution.	white precipitate insoluble in NaOH solution.	Mg^{2+} is confirmed.
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Result:

The given salt contains SO_4^{2-} as acid radical and Mg^{2+} as base radical.
Hence the given salt is MgSO_4 .